

Shubham Kukreja

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Research Interests: Quantum criticality in metals, non-Fermi liquids, renormalization group methods, superconducting instabilities, emergent symmetries, and scale invariance in strongly correlated systems.

Education

- 2023*-26[†] **Ph.D.**, Physics, McMaster University
Thesis Advisor: Dr. Sung-Sik Lee | Research Area: Condensed-matter theory
- 2021-22[‡] **M.Sc.**, Physics, McMaster University
Thesis Advisor: Dr. Sung-Sik Lee | Research Area: Condensed-matter theory
GPA: 3.96/4.0; Transferred to Ph.D. program
- 2018-21 **B.Sc.**, Physics (with Co-op), University of Windsor
Double Minor in Mathematics and Computer Science
Research Advisors: Dr. Chitra Rangan and Dr. Jeffrey Rau
Graduated with Great Distinction, Outstanding Scholar Medalist
Cumulative Average: 96.5%, Major Average: 97.2%

Research Experience

- 2021-Present Doctoral Researcher, McMaster University, (Advisor: Dr. Sung-Sik Lee)
- Developed a functional renormalization group (fRG) framework to construct the full low-energy effective field theory of non-Fermi liquids with hot Fermi surfaces, going beyond conventional patch-based descriptions.
 - Introduced *projective fixed points* (PFPs) as the organizing structures of renormalization group flows in strongly coupled metallic systems, and used the topology of families of PFPs to classify non-Fermi-liquid phases into universality classes.
 - Derived experimentally testable predictions across universality classes, including critical exponents, superconducting transition temperatures, pairing instabilities, and emergent symmetries dictated by the projective nature of the fixed points.
 - Established a distinct form of scale invariance in metals, termed *rational scale invariance*, by analyzing angle-resolved pairing susceptibilities in non-Fermi liquids with hot Fermi surfaces, in contrast to conventional continuous scale invariance.
- 2020-21 Undergraduate Researcher, University of Windsor, (Advisors: Dr. Chitra Rangan and Dr. Jeffrey Rau)
- Simulated [High Harmonic Generation from monolayer graphene](#) by studying the non-linear response and optimizing field parameters
- 2019-21 Undergraduate Researcher, Institute for Diagnostic Imaging Research, (Advisor: Dr. Roman Maev)
- Developed physics-informed ultrasonic signal-processing and feature-engineering pipelines for machine-learning-based, real-time non-destructive evaluation of resistance spot welds, including A-scan analysis and automated dataset generation.

*January 2023

*Expected

‡December 2022

Publications

 [Google Scholar](#)

Preprints and Journal Articles

- **Kukreja, S.,** Willerton, D. M. & Lee, S.-S. Classification of non-Fermi liquids and universal superconducting fluctuations. arXiv: [2601.21047 \[cond-mat.str-el\]](#) (2026). (submitted to Phys. Rev. B)
 - Comprehensive study (123 pages).
 - Classification of non-Fermi liquids with globally convex hot Fermi surfaces based on projective fixed points and renormalization of the Fermi momentum, including experimentally testable predictions for emergent symmetry, universal scaling behavior and superconducting instabilities.
- **Kukreja, S.,** Besharat, A. & Lee, S.-S. Projective fixed points for non-Fermi liquids: A case study of the Ising-nematic quantum critical metal. *Phys. Rev. B* **110**, 155142. <https://link.aps.org/doi/10.1103/PhysRevB.110.155142> (15 Oct. 2024).
 - Comprehensive study (57 pages).
 - *Editors' Suggestion, Graduate Student Publication Award* (McMaster Faculty of Science)
 - The paper investigates the physical implications of projective fixed points in the renormalization group flow of Ising-nematic quantum critical metals, identifying universality classes characterized by non-tunable forward interaction and intrinsic superconducting fluctuations.

To Be Submitted

- **Kukreja, S.,** Willerton, D. M. & Lee, S.-S. Lack of scale invariance of the angle-resolved pair susceptibility as a diagnostic of non-Fermi liquids.
 - The paper demonstrates the emergence of rational scale invariance, rather than continuous scale invariance, in metals with abundant gapless modes by investigating the pairing susceptibility in a class of non-Fermi liquids with hot Fermi surfaces.

Awards & Honors

2026	Dr. Ron and Helene Goodman Physics Award
2026	Yates Scholarship
2025	International Student Award
2025	Mackenzie King Memorial Open Scholarship – Institutional Nominee <i>Nominated by McMaster University as one of two graduate students for national-level competition</i>
2025	Desmond G. Burns Graduate Scholarship (\$4,000)
2024	Graduate Student Publication Award
2021	Ontario Graduate Scholarship (\$15,000)
2020	Research Internship Award (\$6,000)
2018-21	President's Honor Roll and Dean's List
2018-21	Outstanding Scholars (\$9,000)
2019, 2021	Elizabeth Ellen-Tilston Memorial Fund (\$2,000)
2020	Pernod Ricard/Hiram Walker and Sons Scholarship (\$1,000)

Technical Skills

Programming Languages: Python, Mathematica C++, MATLAB

Tools & Software: LaTeX, [Git](#), NumPy, Matplotlib, SciPy

Presentations

Talks

- T1. **Kukreja, S.,** Willerton, D. & Lee, S.-S. *Classification of non-Fermi liquids and their universal superconducting fluctuations: Part II.* [APS March Meeting](#) (Denver, USA). Mar. 2026.

- T2. **Kukreja, S.**, Willerton, D. & Lee, S.-S. *Classification of non-Fermi liquids and universal superconducting fluctuations*. Perimeter Institute Graduate Students Conference (Waterloo, CA). Oct. 2025.
- T3. **Kukreja, S.**, Besharat, A. & Lee, S.-S. *Exploring the landscape of non-Fermi liquids*. Perimeter Institute Graduate Students Conference (Waterloo, CA). Sept. 2024.
- T4. **Kukreja, S.**, Besharat, A. & Lee, S.-S. *Low-energy effective theories of metals*. APCTP Focus Program on Integrability, Duality, and Related Topics (Asian Pacific Center for Theoretical Physics, Virtual). Oct. 2024. (Presented by Lee).
- T5. **Kukreja, S.**, Besharat, A. & Lee, S.-S. *Low-energy effective theories of metals*. 12th Conference on the Exact Renormalization Group (SwissMAP Research Station, Les Diablerets, Switzerland). Sept. 2024. (Presented by Lee).
- T6. **Kukreja, S.**, Besharat, A. & Lee, S.-S. *Non-Fermi liquid theories*. 2024 Strongly Correlated Physics (National Center for Theoretical Science, Taiwan). Aug. 2024. (Presented by Lee).
- T7. **Kukreja, S.**, Besharat, A. & Lee, S.-S. *Space of non-Fermi liquids*. KITP Program: Correlated Gapless Quantum Matter (Santa Barbara, USA). May 2024. (Presented by Lee).
- T8. **Kukreja, S.**, Rau, J. & Rangan, C. *Calculation of High Harmonic Spectrum from a 1D periodic potential*. UWill Discover Conference (Virtual). Mar. 2021.
- T9. **Kukreja, S.**, Rau, J. & Rangan, C. *High Harmonic Generation from 1D periodic potentials*. Canadian Association of Physicists Conference (Virtual). June 2021.
- T10. **Kukreja, S.**, Rau, J. & Rangan, C. *Investigation of the Crank-Nicolson and Imaginary-Time Methods to solve the Time-Dependent Schrödinger Equation for a Double-Well Potential*. Canadian Undergraduate Physics Conference (Virtual). Nov. 2020.

Posters

- P1. **Kukreja, S.**, Chertov, A. & Maev, R. *Real-Time Integrated Weld Analyzer*. American Physical Society Conference (Virtual). Oct. 2020.

Teaching

Teaching Assistant

McMaster University

F* 2021-24	Introductory Mechanics (PHYS 1D03)
F 2024-25	Physics Drop-in
W [†] 2022-26	Waves, Electricity and Magnetism (PHYS 1E03)
W 2022-25	Introductory Physics (PHYS 1A03)
F 2022	Electromagnetic Theory (PHYS 4B03)

University of Windsor

2019-20	Introductory Physics 2 (PHYS 1410)
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Tutor

McMaster University

2022-23	Introductory Physics (PHYS 1A03) & Introduction to Modern Physics (PHYS 1AA3)
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*Fall.

[†]Winter.

Academic Service

2020-21	Mentor , My Sci Mentorship Program, Mentee: Ehsan Salimi (Program: Computer Science), Science Society, University of Windsor
2020	Volunteer , Organization Team, Quantum Holography Workshop, Department of Physics, University of Windsor

References

- Dr. Sung-Sik Lee
Professor, McMaster University and Perimeter Institute
slee@mcmaster.ca
 - Dr. Clifford Burgess
Professor, McMaster University and Perimeter Institute
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 - Dr. Sergey Sibiryakov
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